

Satisfied, Not Solved

What a fairness fix actually does to real decisions

Dheirav

Findings summary — August 13, 2026

Terminology. Technical names are given in *italics* the first time they appear, immediately after the plain description. A full glossary is in `00-plain-english.pdf`.

1 The problem

A program decides who gets a mortgage. It approves **79%** of one group of applicants and **57%** of another. That gap is what people mean by algorithmic bias.

The standard fix is to instruct the program: *approve both groups at the same rate (demographic parity)*. It works — on the standard benchmark dataset the gap between the two rates falls from 0.186 to 0.018, for less than two percentage points of accuracy.

But there are two completely different ways for a program to obey that instruction:

1. Approve *more* people from the group being turned down.
2. Turn down *more* people from the group getting through.

Option 2 makes the numbers equal without helping anybody. It has a name in the ethics literature: *levelling down*.

The score that certifies the fix cannot tell these apart. It measures only whether the two rates match — not how they came to match, and not whether the total number of approvals went up or down.

2 What was already known, and what was not

Known. A 2023 paper by Mittelstadt, Wachter and Russell showed that these fixes mostly do option 2, and argued it is the normal case rather than bad luck. It is in their title: *levelling down “by default”*.

Also known, and I got this wrong at first. That same paper *also* proposes the remedy — their §6 requires that every group receives at least some minimum approval rate. I originally believed my version of this was new. It is not. I found that by reading their paper properly, and corrected my write-up. What is different about mine is narrower: it works without the program ever being told which group an applicant belongs to, and it is tested on data the programs had never seen, across 26 datasets — where they report one dataset, scored on the same data they trained on.

Not known: *when it happens*. Their paper never discusses how generous the system is overall. Their only remark near it is that the benchmark dataset is “more than 75% negatively labelled” — said in passing about accuracy cost, never connected to which of the two options you get. They were working in exactly the regime where their conclusion holds, and did not ask whether it would reverse elsewhere.

The nearest miss. A 2024 paper by Goethals, Delaney, Mittelstadt and Russell studies how the cost of fairness changes with the available budget, and calls this “a factor overlooked in previous evaluations”. Two of those four authors also wrote the 2023 paper. But in their setup the number of approvals is *fixed in advance* as a budget, so the total cannot move and the effect reported here is invisible by construction. The group best placed to find this looked at the same dimension and did not make the connection.

3 The main finding

Levelling down is not what these fixes do. It is what they do when the system is stingy.

Count what fraction of *everyone* gets approved — both groups together. Call it the **approval rate** (*selection rate*).

- Approval rate below roughly **0.3**: the fix takes approvals away.
- Approval rate above roughly **0.6**: the fix hands approvals out.
- In between, it flips.

3.1 How this was shown

Two datasets always differ in dozens of ways, so comparing them proves nothing. Instead, **one dataset was used and exactly one thing was changed.**

The dataset predicts whether someone earns more than \$50,000. That figure is an arbitrary choice by whoever built the benchmark. Raise it and fewer people qualify; lower it and most do. **Same people, same information about them, same groups — only the finish line moves.** Automated test code verifies that the people, the features and the groups really are identical between runs.

Income cutoff	Approval rate	Change in total approvals	Lost per one gained
\$100,000	0.030	−29.71%	22.03
\$70,000	0.099	−22.05%	2.18
\$50,000	0.252	−2.34%	1.14
\$30,000	0.598	+0.98%	0.83
\$20,000	0.760	+0.80%	0.75
\$10,000	0.890	+0.08%	0.89

The last column is how many people lost an approval for each person who gained one. At the top of the table, 22 people lose out for every one who benefits. At the bottom, more people gain than lose.

3.2 It holds up everywhere it was checked

Check	Result
A second population	Same pattern, more strongly
A different way of grouping people	Four populations, all agree
Fourteen populations run <i>after</i> the prediction was written down	All consistent; correlation +0.901
Real mortgage decisions, where no arbitrary cutoff exists	Lands exactly where predicted
Two datasets it was never tuned on	Both placed on the correct side

3.3 The obvious objection, answered in advance

Someone will say: this is just the well-known fact that the two *groups* have different underlying rates.

It is not, and the difference matters. That objection is about a gap *between two groups*. This finding is about the *overall level* for everybody together — a different quantity.

The two do move together, unavoidably: if almost nobody or almost everybody is approved, there is no room for a gap between groups. So the analysis **holds the between-group gap fixed and re-checks**, and the relationship gets *stronger*, not weaker ($+0.801 \rightarrow +0.980$ in one population, $+0.964 \rightarrow +0.994$ in another). That check is the core of the argument, not a footnote to it.

4 Exactly what levelling down is

There is a small piece of arithmetic that makes the whole thing precise, and it appears not to be written down anywhere.

The fix puts both groups on one shared approval rate. Before the fix, the overall rate was just the average of the two groups' rates, weighted by how many people are in each. So:

The total number of approvals is unchanged **exactly when** the shared rate lands at that size-weighted average.

Levelling down is nothing more mysterious than the program picking a shared rate *below* that point. This needs no assumptions and held to within 0.076 percentage points across all fourteen datasets that were run only after this was written down.

5 What I could not work out

Why the flip happens. I derived a candidate explanation from theory, wrote the prediction and its pass mark down in advance, and tested it on fourteen datasets run afterwards. It passed the test I set — and was then beaten by a far cruder rule (“approval rate above one half? then it's the good kind”). So the derivation added nothing and is reported as a failure.

The lesson is recorded with it: **I set a pass mark but never set a rival to beat**, so passing did not mean what it looked like it meant.

How big the effect is. The approval rate tells you the direction, not the size. One population loses 2.34% where another loses 20.5% at almost the same approval rate. A candidate explanation for that was tested and refuted.

6 Supporting findings

Combinations of groups get missed. A fix applied to one characteristic leaves people at the *intersection* of two (say, Black women) worse off than any group the fix was told about — and worse than the benchmark dataset suggests ($9.0\times$ against $13.2\times$).

The standard explainability tool does not reveal what happened. A widely used method for showing which information a program relied on reports a +151% shift after the fix. Three explanations were proposed and none survived; most of the apparent shift turns out to be an accounting artifact between two near-identical pieces of information. **The transferable lesson: this tool can move enormously without the program's behaviour changing much.**

On small datasets the fix is drowned by chance. Below roughly 2,500 test cases, re-running with a different random split moves the numbers more than the fix itself does.

The recommended alternative is worse. *Minimax group fairness* is proposed in the literature as the answer to levelling down. On mortgage data it destroyed 10% of all approvals, at 46.7 lost per one gained — on the very data where the ordinary fix *created* approvals. Its objective minimises the worst group’s *error rate*, and the group with the most errors is not the group being treated worst.

7 Why any of this matters

Loans. Hiring. University admissions.

These are where such fixes are actually deployed, and they are nearly all stingy systems — few applicants are approved.

**So the normal case in real deployments is the harmful one,
and the fairness score never says so.**

8 How carefully this was done

Every result is the average of five runs with different random splits. The program is never allowed to see the protected characteristic. All measures are written from their definitions and checked against a standard library on every run.

Where a result could have gone either way, the prediction and its pass mark were written and committed *before* the experiment ran, and the order is provable from the version history.

That discipline is why the following can be reported honestly: **three of my own predictions failed**; two of my own proposed explanations were disproved by my own experiments; and three claims were corrected after the fact — including my claim that the remedy was new, which the literature had got to first.

9 The question I need answered

Is “we found when this happens” enough on its own for a venue like FAccT or AIES, or does the paper need “and here is why” first?

I can say when the direction flips. I cannot say why — and the attempt is written up as a failure rather than left out.

Every number here is regenerated from stored results by an automated check that fails if a document and its data disagree. The cited papers are in **papers/**; the full research record is 16 documents in **research/docs/**.